

Edge Task Hub

Resource-aware Edge AI automation on Raspberry Pi.

- Authors and student IDs will be added before submission.
- Core claim: Feishu is the message channel, but private inference and scheduling decisions stay on the Pi.

Course Focus

The project must defend Edge AI, not only AI output.

- A cloud chatbot can ignore memory, storage, and cold start.
- A Raspberry Pi must choose models under real device limits.
- The report focuses on setup, results, and analysis.

Edge Boundary

Feishu receives commands and delivers messages; the Pi owns the intelligence.

- Local Ollama summarizes news and private documents.
- Local rules keep deterministic fallback behavior.
- Local Isolation Forest monitors device health.

System Architecture

FastAPI, APScheduler, SQLite, Ollama, and Feishu are joined into one edge workflow.

- Browser UI configures tasks.
- APScheduler runs reminders and RSS digests.
- SQLite stores tasks, runs, events, document summaries, and metrics.
- Feishu receives results and sends commands/files.

Daily News Workflow

Current daily task: 0 10 * * * local time.

- Fetch previous-day RSS items.
- Summarize locally with qwen3:1.7b.
- Send the final summary through Feishu.
- Generation time can be longer because daily summaries do not need strict live latency.

Reminder Workflow

Users can create reminders through Feishu text commands.

- Example: remind me every day at 23:30 to sleep.
- The parser creates a Task row and registers it with the scheduler.
- Built-in lunch and sleep reminders are already seeded.

Private Document Workflow

A private Word or text document can be summarized on the Pi.

- Feishu sends the file event.
- The Pi downloads to a temporary directory.
- Text is extracted locally and summarized with qwen3:1.7b.
- The temporary file is deleted after processing.

Model Exploration Logic

The project moved from larger model quality to edge-device feasibility.

- 2B-size model: better language potential, but unstable installation and storage pressure.
- 0.8B quantized model: smaller, but still slow and empty or length-limited.
- 0.5B model: fast, but weaker summary quality.
- Final policy: choose model by edge job, not by size alone.

Model Results

The final choice is quality-first for waitable summaries and fast paths for short tasks.

Candidate	Constraint	Decision
qwen3.5:2b	Storage and deployment stability	Rejected before final inference
qwen3:1.7b	Quality-first local summarization	Selected
qwen3:0.6b	Low-latency lightweight prompts	Kept as fast model
qwen3.5:0.8b	Memory pressure and cold-start latency	Rejected
qwen2.5:0.5b-instruct	Quality loss from very small LLM	Rejected for final summaries

Why qwen3.5:0.8b Failed

Quantization helped size, but not enough for this Raspberry Pi news-summary workload.

- Recorded cold-start runs were about 52 to 75 seconds.
- Generation speed was about 2 tokens/s.
- Visible output was empty or length-limited in the benchmark.
- The likely causes were cold loading, CPU speed, memory pressure, and thinking-token budget.

Program Models Matter

Edge AI is not only one large neural model.

- Rules fallback keeps summaries deterministic when Ollama is missing or too slow.
- Isolation Forest scores CPU, memory, disk, temperature, and network metrics.
- These models are smaller, faster, and easier to explain for their jobs.

Results

The system now supports the required Edge AI workflows end to end.

- qwen3:1.7b selected for daily news and private document summaries.
- qwen3:0.6b kept as fast local prompt model.
- Rules fallback and Isolation Forest kept for reliability and local monitoring.
- 10 unit tests pass on the Raspberry Pi.

Discussion

The strongest project argument is the edge design, not only the LLM output.

- Private content stays on the device during summarization.
- Failures are recorded instead of hidden.
- The system chooses different local intelligence for text and metrics.

Limitations

The final defense should be honest about current boundaries.

- Feishu public callback configuration is still required.
- Local LLM cold starts are slow on Raspberry Pi.
- Scanned PDFs need OCR and were not part of the verified path.
- Names and student IDs still need to be inserted.

Conclusion

Edge Task Hub is an Edge AI automation system, not just a scheduler.

- Local summaries, local document processing, local anomaly detection.
- Feishu-based user interaction.
- Resource-aware model choice with qwen3:1.7b as the best summary model.